Software Requirements Specification (SRS)

Demand Forecasting & Inventory Planning System

Project	Demand Forecasting & Inventory Planning System
Version	1.1
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Revision History

Version	Date	Author	Change Summary
1.0	07-10-2026	Team 03	Initial SRS tailored to assigned mini-project
1.1	12-10-2026	Team 03	Consistency fixes; role-permission matrix; DFIP-F-023; forecasting/planning rules; CSV format; RTM alignment.

Approvals

Role	Name	Signature / Email	Date
Course Instructor			
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1. Introduction

1.1 Purpose

This document is the Software Requirements Specification (SRS) for the Demand Forecasting & Inventory Planning System. It defines the functional requirements, non-functional requirements, external interfaces, security requirements, acceptance criteria, system models, and requirement traceability for the Team 03 Software Engineering mini-project.

1.2 Scope

The system uses historical sales and current inventory data to estimate future product demand, recommend reorder quantities, and identify stockout and overstock risks. It provides inventory and sales data management, demand forecasting, planning dashboards, alerts, and downloadable reports. The system supports decision-making for inventory planners and administrators; it does not directly place purchase orders with suppliers or control physical warehouse equipment in the initial version.

1.3 Audience

The intended audience includes the project team, course instructors and evaluators, developers, testers, inventory planners, administrators, and future maintainers of the system.

1.4 Definitions

Term	Meaning
DFIP	Demand Forecasting & Inventory Planning
SKU	Stock Keeping Unit
Forecast Horizon	Future period for which demand is predicted
Reorder Quantity	Recommended amount of stock to replenish
Stockout	Situation in which available inventory is insufficient to meet demand
Overstock	Inventory held substantially above expected demand
API	Application Programming Interface
RTM	Requirements Traceability Matrix

2. Overall Description

2.1 Product Perspective

The product is a data-driven inventory decision-support system. Historical sales records and current stock data are stored in a central database. The forecasting module analyzes past demand patterns and produces future demand estimates. The planning module compares the predicted demand with available stock to recommend replenishment quantities and classify stockout or overstock risk. Users interact through a web-based dashboard.

2.2 Major Product Functions (Detailed)

- Authenticate users and provide role-based access.
- Manage product/SKU details and inventory thresholds.
- Enter or import historical sales data.
- Record and update current inventory levels.
- Generate future demand forecasts for selected products and time periods.

- Compare predicted demand against available inventory.
- Recommend reorder quantities.
- Identify potential stockout and overstock risks.
- Present forecasts, stock status, trends, and alerts on dashboards.
- Generate and export planning reports.
- Maintain configuration values and audit important user actions.

2.3 User Roles and Characteristics

Role	Description	Typical Needs
Administrator	Maintains users, products, system settings and data access.	Reliable administration, validation, auditability.
Inventory Manager / Planner	Reviews inventory, forecasts and reorder suggestions.	Clear forecasts, warnings and actionable reorder information.
Viewer / Management User	Reviews dashboard and reports without changing operational data.	Simple summaries and downloadable reports.

2.3.1 Role-Permission Matrix

Function	Administrator	Planner	Viewer
Login / logout	Allowed	Allowed	Allowed
View dashboard, forecasts, risks and reports	Allowed	Allowed	View only
Manage products / SKUs	Allowed	View only	View only
Update inventory quantities	Allowed	Allowed	Not allowed
Add / import sales data	Allowed	Allowed	Not allowed
Generate forecasts and planning recommendations	Allowed	Allowed	Not allowed
Export reports	Allowed	Allowed	Allowed
Manage users and roles	Allowed	Not allowed	Not allowed
Configure planning parameters	Allowed	Not allowed	Not allowed
View audit records	Allowed	Not allowed	Not allowed

Permissions are enforced on the server side. The Viewer role is read-only and cannot change operational or administrative data.

2.4 Operating Environment

- Modern web browser (Chrome/Edge/Firefox).
- Application server running on Windows or Linux.
- Relational database such as MySQL/PostgreSQL.
- Python-based forecasting service or equivalent analytics module.
- Local network or Internet connectivity for web access.

2.5 Constraints

- Forecast quality depends on the completeness and quality of historical sales data.
- The first version is intended for academic mini-project scale and moderate data volumes.
- Only authorized users may modify inventory, sales, or configuration data.
- The system must preserve traceability between requirements, design components, and test cases.
- External supplier ordering/payment integration is outside the initial scope.

3. External Interface Requirements

3.1 User Interfaces

- Login page for authorized access.
- Dashboard showing inventory health, forecasts, reorder recommendations, and risk alerts.
- Forms for product, sales, and inventory data entry.
- Forecast screen with product selection, forecast horizon, predicted values, and trends.
- Reports screen with filters and export option.
- Administrative screens for users and system configuration.

3.2 Hardware Interfaces

No specialized hardware is mandatory. The application runs on standard desktop/laptop systems and can be hosted on a local or cloud server. Barcode scanners or warehouse devices are not required in the initial version.

3.3 Software Interfaces

- Relational database for users, products, sales, inventory, forecasts, recommendations, and audit data.
- Forecasting/analytics module for demand prediction.
- CSV file import/export interface for historical sales and reports.
- Optional REST API interface between web frontend and backend services.

3.4 Communications

- HTTP/HTTPS for browser-to-server communication.
- JSON for REST API request and response payloads where APIs are used.
- TLS/HTTPS should be used when the system is deployed over a network.

Template compliance target: the project defines at least 15 functional requirements, 5 non-functional requirements, 2 security objectives, and 5 security requirements.

4. System Features (Detailed)

4.1 Authentication & Access Control

Description: Controls user sign-in and authorized access.

Req ID	Requirement (shall…)	Type	Priority	Source / Stakeholder	Acceptance Criteria	Test Case Ref
DFIP-F-001	The system shall authenticate registered users using a valid username/email and password.	Functional	High	Security / Users	Valid credentials open an authorized session; invalid credentials are rejected.	TC-AUTH-01
DFIP-F-002	The system shall enforce role-based access so that only	Functional	High	Security / Admin	Unauthorized roles cannot access administrator-only functions.	TC-AUTH-02

	administrators can perform administrative operations.					
DFIP-F-003	The system shall allow an authenticated user to log out and terminate the active session.	Functional	Medium	Users	After logout, protected pages require authentication again.	TC-AUTH-03

4.2 Product & Inventory Management

Description: Maintains the product catalog and stock information.

Req ID	Requirement (shall…)	Type	Priority	Source / Stakeholder	Acceptance Criteria	Test Case Ref
DFIP-F-004	The system shall allow an administrator to create, view, update, and deactivate product/SKU records.	Functional	High	Admin / Inventory	Product changes are saved and visible in the product list.	TC-PROD-01
DFIP-F-005	The system shall store current inventory quantity and reorder-related parameters for each active product.	Functional	High	Inventory	Saved inventory values are retrievable for the selected product.	TC-INV-01
DFIP-F-006	The system shall allow authorized users to update inventory quantities with validation against negative or invalid values.	Functional	High	Inventory	Invalid quantities are rejected; valid values update stock.	TC-INV-02

4.3 Sales Data Management

Description: Captures historical demand information used by forecasting.

Req ID	Requirement (shall…)	Type	Priority	Source / Stakeholder	Acceptance Criteria	Test Case Ref
DFIP-F-007	The system shall allow authorized users to add historical sales records with product, date/period, and quantity sold.	Functional	High	Inventory / Data	Valid sales records are stored with correct product and period.	TC-SALES-01
DFIP-F-008	The system shall support	Functional	High	Data	Valid rows are imported and	TC-SALES-02

	importing historical sales data from a UTF-8 CSV file with columns sku,date,quantity_sold, where date uses YYYY-MM-DD format.				invalid rows are reported; malformed files are rejected without corrupting stored data.	
DFIP-F-009	The system shall allow users to view and filter historical sales by product and date range.	Functional	Medium	Users	Filtered results match the selected product and date range.	TC-SALES-03

4.4 Demand Forecasting

Description: Produces future demand estimates from historical sales.

Req ID	Requirement (shall…)	Type	Priority	Source / Stakeholder	Acceptance Criteria	Test Case Ref
DFIP-F-010	The system shall generate a future demand forecast for a selected product using its available historical sales data.	Functional	High	Business / Planner	A forecast is generated only when at least 8 valid weekly/monthly periods are available.	TC-FCST-01
DFIP-F-011	The system shall allow the user to select a supported forecast horizon of 1-12 weeks or 1-6 months.	Functional	High	Planner	Forecast output contains exactly the requested number of consecutive weekly or monthly points.	TC-FCST-02
DFIP-F-012	The system shall store generated forecast results with product, forecast period, predicted demand, model/version information, and generation timestamp.	Functional	Medium	Audit / Analytics	Stored forecast can be retrieved with model name, model version, generation timestamp, period and predicted quantity metadata.	TC-FCST-03

4.4.1 Forecasting Rules

Historical sales are validated, aggregated to weekly or monthly periods, missing periods are filled with zero, and duplicate dates use the last value. At least 8 periods are required. Holt-Winters exponential smoothing is used when at least two seasonal cycles are available; otherwise simple exponential smoothing is used. Negative predictions are clipped to 0. Supported horizons are 1-12 weeks or 1-6 months.

4.5 Inventory Planning & Risk Identification

Description: Converts demand forecasts into inventory decisions.

Req ID	Requirement (shall…)	Type	Priority	Source / Stakeholder	Acceptance Criteria	Test Case Ref
DFIP-F-013	The system shall calculate reorder quantity as max(0, ceil(forecast_demand + safety_stock - current_stock)).	Functional	High	Inventory Planner	Recommendation is generated from forecast demand, current stock and configured safety stock; e.g., demand 100, safety 20, stock 50 gives reorder 70.	TC-PLAN-01
DFIP-F-014	The system shall classify stockout risk when current_stock < forecast_demand.	Functional	High	Inventory Planner	Product is flagged as stockout when current stock is below forecast demand.	TC-PLAN-02
DFIP-F-015	The system shall classify overstock risk when current_stock > overstock_factor x forecast_demand.	Functional	High	Inventory Planner	Product is flagged as overstock when stock exceeds the configured overstock threshold; otherwise risk is none.	TC-PLAN-03
DFIP-F-016	The system shall allow an authorized user to review the inputs used for each reorder recommendation.	Functional	Medium	Planner / Audit	User can view forecast, stock, and planning parameters associated with the recommendation.	TC-PLAN-04

4.5.1 Planning Rules

Risk classification is deterministic: stockout if current_stock < forecast_demand; overstock if current_stock > overstock_factor x forecast_demand; otherwise none. The default overstock factor in the data model is 1.5, and every recommendation retains the inputs used for traceability.

4.6 Dashboard, Alerts & Reports

Description: Communicates planning results to users.

Req ID	Requirement (shall…)	Type	Priority	Source / Stakeholder	Acceptance Criteria	Test Case Ref
DFIP-F-017	The system shall display a dashboard summarizing current stock, predicted demand, reorder recommendations, and risk counts.	Functional	High	Management / Planner	Dashboard shows current values from the latest stored data.	TC-DASH-01

DFIP-F-018	The system shall display alerts for products currently classified as stockout or overstock risk.	Functional	High	Planner	Risk products appear in the alerts view with risk type.	TC-DASH-02
DFIP-F-019	The system shall generate inventory planning reports filtered by product and/or time period.	Functional	Medium	Management	Report contains requested forecast and inventory-planning information.	TC-RPT-01
DFIP-F-020	The system shall allow authorized users to export supported reports in CSV format.	Functional	Medium	Users	Downloaded CSV contains the same report data shown in the system.	TC-RPT-02

4.7 Administration & Audit

Description: Supports configuration and traceability.

Req ID	Requirement (shall…)	Type	Priority	Source / Stakeholder	Acceptance Criteria	Test Case Ref
DFIP-F-021	The system shall allow an administrator to manage user accounts and assigned roles.	Functional	High	Admin	User/role changes persist and affect access permissions.	TC-ADM-01
DFIP-F-022	The system shall maintain audit records for security-sensitive and data-changing operations such as login, product updates, inventory updates, imports, and configuration changes.	Functional	Medium	Security / Audit	Audit record stores actor, action, date/time, and result.	TC-AUD-01
DFIP-F-023	The system shall allow an administrator to configure forecast/planning parameters including default safety stock, overstock factor, and minimum history.	Functional	Medium	Admin / Planning	Saved parameter values are retrievable and are applied to subsequent forecast/planning operations.	TC-CFG-01

5. Non-Functional Requirements (Detailed)

Req ID	Requirement	Category	Priority	Acceptance Criteria / Measurement	Test Ref
DFIP-NF-001	For normal academic-scale usage, dashboard and standard data retrieval operations shall complete within 3 seconds for at least 90% of requests.	Performance	High	90th percentile response time <= 3 seconds in the defined test environment.	TC-PERF-01
DFIP-NF-002	A demand forecast for a single product and supported horizon shall complete within 10 seconds under the defined test dataset.	Performance	High	Forecast generation time <= 10 seconds.	TC-PERF-02
DFIP-NF-003	The system shall preserve committed product, sales, inventory, forecast, and audit data across normal application restarts.	Reliability	High	Restart test shows committed data remains available and consistent.	TC-REL-01
DFIP-NF-004	The web interface shall be usable on common desktop screen sizes and shall provide clear validation/error messages for user input.	Usability	Medium	Usability checklist passes; validation messages are visible and understandable.	TC-UX-01
DFIP-NF-005	The codebase shall use modular components with documented interfaces so forecasting, inventory, and reporting modules can be maintained independently.	Maintainability	Medium	Design review confirms modular separation and documented interfaces.	TC-MAINT-01

5.1 Security

5.1.1 Security Objectives

- SO-1 - Protect the confidentiality and integrity of user credentials, sales, inventory, forecast, and planning data.
- SO-2 - Ensure that only authenticated and authorized users can access or modify functions and data according to their assigned roles.

5.1.2 Security Requirements

Req ID	Requirement (shall…)	Type	Priority	Acceptance Criteria	Test Ref
DFIP-SR-001	The system shall store user passwords using a secure one-way password-hashing mechanism rather than plaintext.	Security	High	Database inspection shows no plaintext passwords; authentication still succeeds with valid credentials.	TC-SEC-01
DFIP-SR-002	The system shall require authentication before access to protected application pages or APIs.	Security	High	Unauthenticated requests to protected resources are rejected or redirected to login.	TC-SEC-02
DFIP-SR-003	The system shall authorize restricted operations according to the user role on the server side.	Security	High	Non-admin role cannot execute admin operation even through direct request.	TC-SEC-03
DFIP-SR-004	The system shall validate and sanitize user-supplied input and imported data before processing or storing it.	Security	High	Malformed/invalid values are rejected without corrupting data or executing unintended commands.	TC-SEC-04
DFIP-SR-005	When deployed over a network, the system shall support HTTPS/TLS for browser and API communication.	Security	Medium	Deployment verification confirms HTTPS endpoint and protected credentials in transit.	TC-SEC-05

6. Quality Attributes & Acceptance Tests

- Correctness: All high-priority functional requirements must meet their stated acceptance criteria.
- Performance: DFIP-NF-001 and DFIP-NF-002 targets must be met in the agreed test environment.
- Reliability: No loss of committed data during normal application restart testing.
- Security: All five defined security requirements must pass their associated security validation tests.
- Traceability: Every listed requirement must map to at least one design/module reference and test case in the RTM.
- Exit criteria for acceptance: all high-priority requirements implemented and verified, no open critical defects, and planned acceptance tests completed.

7. System Models and Diagrams

7.1 UML Use-Case Diagram 1 - Inventory & Sales Management

Figure 1 shows how the Inventory Manager and Administrator interact with authentication, product data, historical sales, inventory updates, dashboard information, and inventory-risk alerts.

7.2 UML Use-Case Diagram 2 - Forecasting & Planning

Figure 2 focuses on forecast generation, forecast review, reorder recommendations, risk analysis, report export, and administrator configuration of forecasting/planning parameters.

8. Requirements Traceability Matrix (RTM)

Req ID	Requirement Short	Section Ref	Module	Test Case(s)	Status (N/P/A)	Comments
DFIP-F-001	The system shall authenticate registered users using a ...	Section 4	Authentication & RBAC Component	TC-AUTH-01	N	
DFIP-F-002	The system shall enforce role-based access so that only...	Section 4	Authentication & RBAC Component	TC-AUTH-02	N	
DFIP-F-003	The system shall allow an authenticated user to log out...	Section 4	Authentication & RBAC Component	TC-AUTH-03	N	
DFIP-F-004	The system shall allow an administrator to create, view...	Section 4	Product Management Component	TC-PROD-01	N	
DFIP-F-005	The system shall store current inventory quantity and r...	Section 4	Inventory Management Component	TC-INV-01	N	
DFIP-F-006	The system shall allow authorized users to update inven...	Section 4	Inventory Management Component	TC-INV-02	N	
DFIP-F-007	The system shall allow authorized users to add historic...	Section 4	Sales Data Component	TC-SALES-01	N	
DFIP-F-008	The system shall support importing historical sales dat...	Section 4	Sales Data Component	TC-SALES-02	N	
DFIP-F-009	The system shall allow users to view and filter histori...	Section 4	Sales Data Component	TC-SALES-03	N	
DFIP-F-010	The system shall generate a future demand forecast for ...	Section 4	Forecasting Engine	TC-FCST-01	N	
DFIP-F-011	The system shall allow the user to select a supported f...	Section 4	Forecasting Engine	TC-FCST-02	N	

DFIP-F-012	The system shall store generated forecast results with ...	Section 4	Forecasting Engine	TC-FCST-03	N	
DFIP-F-013	The system shall calculate a recommended reorder quanti...	Section 4	Planning & Risk Component	TC-PLAN-01	N	
DFIP-F-014	The system shall identify a stockout risk when projecte...	Section 4	Planning & Risk Component	TC-PLAN-02	N	
DFIP-F-015	The system shall identify an overstock risk when invent...	Section 4	Planning & Risk Component	TC-PLAN-03	N	
DFIP-F-016	The system shall allow an authorized user to review the...	Section 4	Planning & Risk Component	TC-PLAN-04	N	
DFIP-F-017	The system shall display a dashboard summarizing curren...	Section 4	Dashboard Component	TC-DASH-01	N	
DFIP-F-018	The system shall display alerts for products currently ...	Section 4	Dashboard Component	TC-DASH-02	N	
DFIP-F-019	The system shall generate inventory planning reports fi...	Section 4	Reporting Component	TC-RPT-01	N	
DFIP-F-020	The system shall allow authorized users to export suppo...	Section 4	Reporting Component	TC-RPT-02	N	
DFIP-F-021	The system shall allow an administrator to manage user ...	Section 4	Administration Component	TC-ADM-01	N	
DFIP-F-022	The system shall maintain audit records for security-se...	Section 4	Audit Component	TC-AUD-01	N	
DFIP-F-023	Administrator configures forecast/planning parameters.	Section 4	Administration Component	TC-CFG-01	N	Added in v1.1
DFIP-NF-001	For normal academic-scale usage, dashboard and standard...	Section 5	Performance / Relevant modules	TC-PERF-01	N	
DFIP-NF-002	A demand forecast for a single product and supported ho...	Section 5	Performance / Relevant modules	TC-PERF-02	N	
DFIP-NF-003	The system shall preserve committed product, sales, inv...	Section 5	Reliability / Relevant modules	TC-REL-01	N	
DFIP-NF-004	The web interface shall be usable on common desktop scr...	Section 5	Usability / Relevant modules	TC-UX-01	N	
DFIP-NF-005	The codebase shall use modular components with	Section 5	Maintainability / Relevant modules	TC-MAINT-01	N	

	document...					
DFIP-SR-001	The system shall store user passwords using a secure on...	Section 5.1	Security / Relevant modules	TC-SEC-01	N	
DFIP-SR-002	The system shall require authentication before access t...	Section 5.1	Security / Relevant modules	TC-SEC-02	N	
DFIP-SR-003	The system shall authorize restricted operations accord...	Section 5.1	Security / Relevant modules	TC-SEC-03	N	
DFIP-SR-004	The system shall validate and sanitize user-supplied in...	Section 5.1	Security / Relevant modules	TC-SEC-04	N	
DFIP-SR-005	When deployed over a network, the system shall support ...	Section 5.1	Security / Relevant modules	TC-SEC-05	N	

Status codes: N = Not yet executed, P = Passed, A = Accepted/Approved.